

Funkcja i jej własności

$$X \times Y = \{x_1, x_2, x_3, x_4\} \times \{y_1, y_2, y_3, y_4\} = \{(x, y) : x \in X, y \in Y\}$$

$$f(x) = x \quad g(x) = \frac{x^2}{x}$$

$$f \neq g, \text{ gdyż } D_f = \mathbb{R}, D_g = \mathbb{R} \setminus \{0\}$$

$$f(x) = \sqrt{\lg^3(2^x + x^2)}$$

$$f'(x) = \frac{3 \lg^2(2^x + x^2)}{2 \sqrt{\lg^3(2^x + x^2)}} \cdot \frac{1}{\cos^2(2^x + x^2)} \cdot (2^x \ln 2 + 2x)$$

Funkcja odwrotna

$$y = 5x - 4 = f(x)$$

$$x = \frac{y+4}{5}$$

$$y = \frac{x+4}{5} = f^{-1}(x)$$

$$f(x) = y = x^2$$

$$D_f = \mathbb{R}$$

$$Z_{Wf} = [0; +\infty)$$

$$x = \sqrt{y} \vee x = -\sqrt{y}$$

$$y = \sqrt{x} \vee y = -\sqrt{x}$$

$$\pi = 180^\circ$$

α	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \alpha$	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1

Tw. funkcja f ma funkcję odwrotną
(f^{-1}) tylko tam gdzie jest różnowartościowa.

Funkcje cyklotometryczne, czyli odwrotne

do trygonometrycznych

$$① f(x) = y = \sin x$$

$$D_f = \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$$

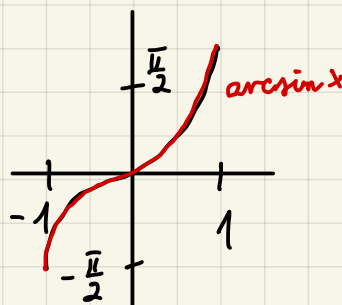
$$Z_{Wf} = [-1; 1]$$

$x =$ "kąt, którego sinus wynosi y " $= \arcsin y$

$$f^{-1}: y = \arcsin x$$

$$D_{f^{-1}} = [-1; 1]$$

$$Z_{Wf^{-1}} = \left[-\frac{\pi}{2}; \frac{\pi}{2}\right]$$



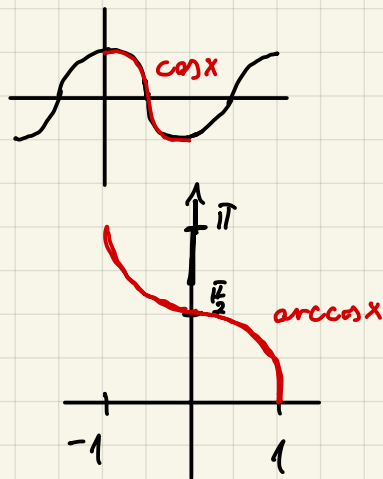
② $y = \cos x$

$D_f = [0; \pi]$

$ZW_f = [-1; 1]$

$x = \arccos y$

$f^{-1}: y = \arccos x$



$\arccos 0 = \frac{\pi}{2}$

$\arccos 1 = 0$

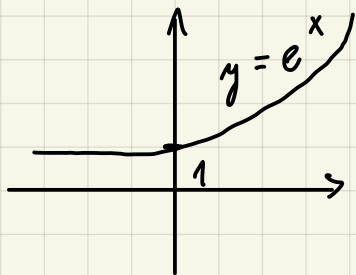
$\arccos(-1) = \pi$

$\arccos(-\frac{1}{2}) = \frac{2}{3}\pi$

$\arccos(-\frac{\sqrt{3}}{2}) = \frac{3}{4}\pi$

$f(x) = e^x = \exp(x)$

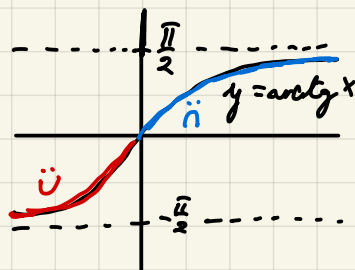
$D_f = \mathbb{R} \quad ZW_f = (0; \infty)$



③ $y = \arctg x$

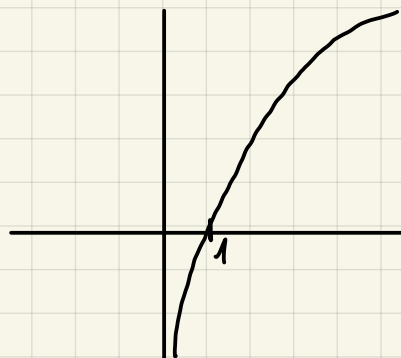
$D = \mathbb{R}$

$ZW = (-\frac{\pi}{2}; \frac{\pi}{2})$



$\log_e x = \ln x = f(x)$
 $e > 1$

$D_f: x > 0 \quad ZW_f = \mathbb{R}$



Derivacje "nowych" funkcji

$$(e^x)' = e^x$$

$$(\ln x)' = \frac{1}{x}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arccos x)' = \frac{-1}{\sqrt{1-x^2}}$$

$$(\operatorname{arctg} x)' = \frac{1}{1+x^2}$$

$$(\operatorname{arcctg} x)' = \frac{-1}{1+x^2}$$

Egzamin !

Tw. Funkcja $f: D_f \rightarrow \mathbb{R}$ jest ciągła w punkcie $x_0 \in D_f$ jeśli

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0} f(x)$$

Dobrać wp. $a \in \mathbb{R}$ tak by dana

funkcja była ciągła w $x_0 = 2$

$$f(x) = \begin{cases} \frac{x^2-4}{x-2} & x \neq 2 \\ a & x = 2 \end{cases}$$

$$\lim_{x \rightarrow 2^+} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2^+} (x+2) = 4 = f(2) = a$$

Odp. dla $a = 4$.

$$\begin{aligned} [\arcsin(\cos(x^2+2x+4))]' &= \\ &= \frac{(\cos(x^2+2x+4))'}{\sqrt{1-\cos^2(x^2+2x+4)}} = \frac{(x^2+2x+4)' \cdot (-\sin(x^2+2x+4))}{\sqrt{\sin^2(x^2+2x+4)}} = \\ &= \frac{(2x+2)(-\sin(x^2+2x+4))}{|\sin(x^2+2x+4)|} \end{aligned}$$

Przykład (granica funkcji złożonej)

$$\lim_{x \rightarrow 0} e^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{1}{x}}$$

$$e^{\lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right) \rightarrow \infty} \rightarrow e^\infty \rightarrow \infty$$

$$e^{\lim_{x \rightarrow 0^-} \frac{1}{x} \rightarrow -\infty} \rightarrow e^{-\infty} = \frac{1}{e^\infty} \rightarrow 0$$